MSG No. 149.24 P_c312 [Type IV, trigonal]

Table 1: Wyckoff site: `2a`, site symmetry: `3.2`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,4,5,6]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[7,8,9,10,11,12]`

Table 2: Wyckoff site: `2b`, site symmetry: `3.2'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,3,10,11,12]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[4,5,6,7,8,9]`

Table 3: Wyckoff site: `2c`, site symmetry: `3.2`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ 0\right]$	`[1,2,3,4,5,6]`
2	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{2}\right]$	`[7,8,9,10,11,12]`

Table 4: Wyckoff site: `2d`, site symmetry: `3.2'`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{1}{4}\right]$	`[1,2,3,10,11,12]`
2	$\left[\frac{1}{3},\ \frac{2}{3},\ \frac{3}{4}\right]$	`[4,5,6,7,8,9]`

Table 5: Wyckoff site: `2e`, site symmetry: `3.2`

No.	position	mapping
1	$\left[\frac{2}{3},\ \frac{1}{3},\ 0\right]$	`[1,2,3,4,5,6]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}\right]$	`[7,8,9,10,11,12]`

Table 6: Wyckoff site: `2f`, site symmetry: `3.2'`

No.	position	mapping
1	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{4}\right]$	`[1,2,3,10,11,12]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{3}{4}\right]$	`[4,5,6,7,8,9]`

Table 7: Wyckoff site: `4g`, site symmetry: `3..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3]`
2	$[0,\ 0,\ -z]$	`[4,5,6]`

continued ...

Table 7

No.	position	mapping
3	$\left[0,\, 0,\, z+\frac{1}{2}\right]$	[7,8,9]
4	$\left[0,\, 0,\, \frac{1}{2}-z\right]$	[10,11,12]

Table 8: Wyckoff site: 4h, site symmetry: 3..

No.	position	mapping
1	$\left[\frac{1}{3},\, \frac{2}{3},\, z\right]$	[1,2,3]
2	$\left[\frac{1}{3},\, \frac{2}{3},\, -z\right]$	[4,5,6]
3	$\left[\frac{1}{3},\, \frac{2}{3},\, z+\frac{1}{2}\right]$	[7,8,9]
4	$\left[\frac{1}{3},\, \frac{2}{3},\, \frac{1}{2}-z\right]$	[10,11,12]

Table 9: Wyckoff site: 4i, site symmetry: 3..

No.	position	mapping
1	$\left[\frac{2}{3},\, \frac{1}{3},\, z\right]$	[1,2,3]
2	$\left[\frac{2}{3},\, \frac{1}{3},\, -z\right]$	[4,5,6]
3	$\left[\frac{2}{3},\, \frac{1}{3},\, z+\frac{1}{2}\right]$	[7,8,9]
4	$\left[\frac{2}{3},\, \frac{1}{3},\, \frac{1}{2}-z\right]$	[10,11,12]

Table 10: Wyckoff site: 6j, site symmetry: ..2

No.	position	mapping
1	$[x,\, -x,\, 0]$	[1,6]
2	$[x,\, 2x,\, 0]$	[2,4]
3	$[-2x,\, -x,\, 0]$	[3,5]
4	$\left[x,\, -x,\, \frac{1}{2}\right]$	[7,12]
5	$\left[x,\, 2x,\, \frac{1}{2}\right]$	[8,10]
6	$\left[-2x,\, -x,\, \frac{1}{2}\right]$	[9,11]

Table 11: Wyckoff site: 6k, site symmetry: ..2'

No.	position	mapping
1	$\left[x,\, -x,\, \frac{1}{4}\right]$	[1,12]
2	$\left[x,\, 2x,\, \frac{1}{4}\right]$	[2,10]
3	$\left[-2x,\, -x,\, \frac{1}{4}\right]$	[3,11]
4	$\left[x,\, 2x,\, \frac{3}{4}\right]$	[4,8]
5	$\left[-2x,\, -x,\, \frac{3}{4}\right]$	[5,9]
6	$\left[x,\, -x,\, \frac{3}{4}\right]$	[6,7]

Table 12: Wyckoff site: `12l`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-y,\ x-y,\ z]$	`[2]`
3	$[-x+y,\ -x,\ z]$	`[3]`
4	$[x,\ x-y,\ -z]$	`[4]`
5	$[-x+y,\ y,\ -z]$	`[5]`
6	$[-y,\ -x,\ -z]$	`[6]`
7	$\left[x,\ y,\ z+\frac{1}{2}\right]$	`[7]`
8	$\left[-y,\ x-y,\ z+\frac{1}{2}\right]$	`[8]`
9	$\left[-x+y,\ -x,\ z+\frac{1}{2}\right]$	`[9]`
10	$\left[x,\ x-y,\ \frac{1}{2}-z\right]$	`[10]`
11	$\left[-x+y,\ y,\ \frac{1}{2}-z\right]$	`[11]`
12	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	`[12]`